

New exotic states from LHCb

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on behalf of LHCb experiment

ORIGINS Cluster

June, 23rd, 2023

1 Introduction

2 Baryons and Pentaquarks

- Ξ_b^- decays
- B decays to $J/\psi \Lambda \bar{p}$
- Update on Ω_c^{*} states (2022-043)
- Update on Ω_c^{*} states

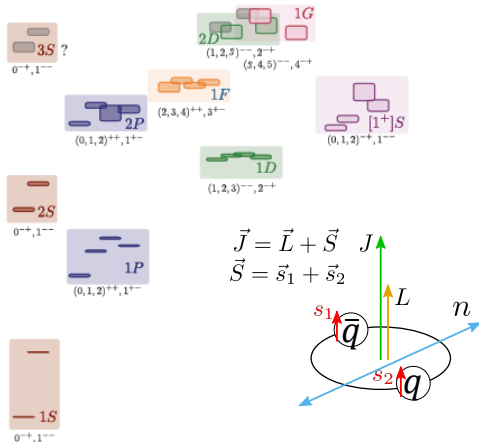
3 Mesons and Tetraquarks in DDK

- $B^+ \rightarrow D^+ D^- K^+$
- $B^+ \rightarrow D_s^+ D_s^- K^+$
- $B^+ \rightarrow D^- D_s^+ \pi^+$ and $B^0 \rightarrow \bar{D}^0 D_s^+ \pi^-$

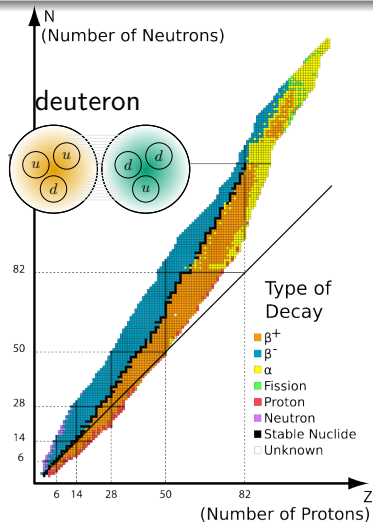
4 Plans

Two ways of building complexity

Hadron variety



Atomic variety

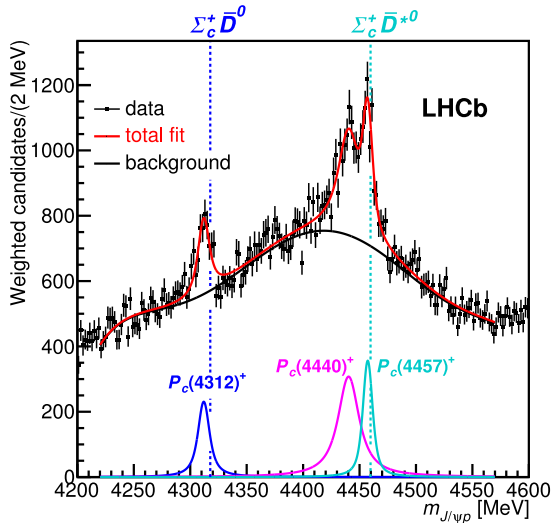
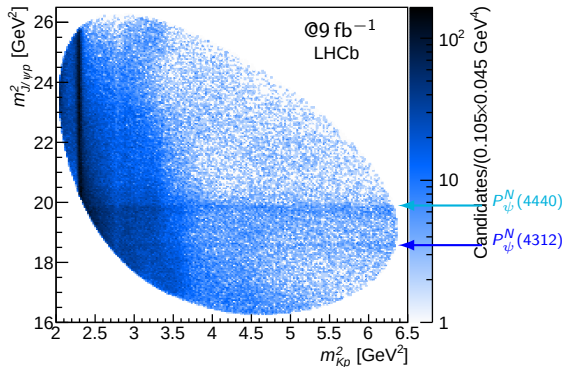


$$\Lambda_b^0 \rightarrow \underbrace{J/\psi p}_{pN_\psi} K^-$$

The first pentaquarks

[PRL 115 (2015), 072001; PRL 122 (2019) 22, 222001]

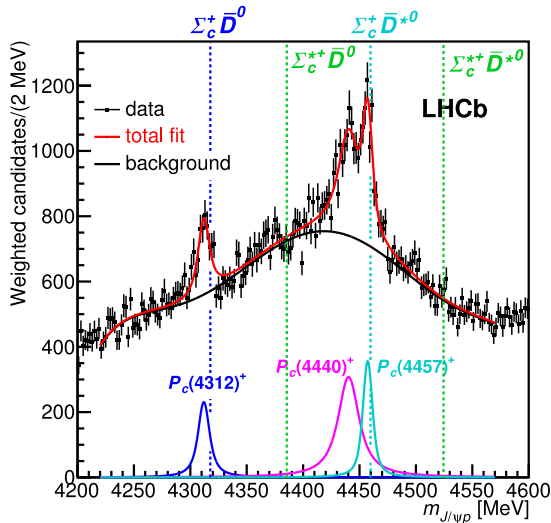
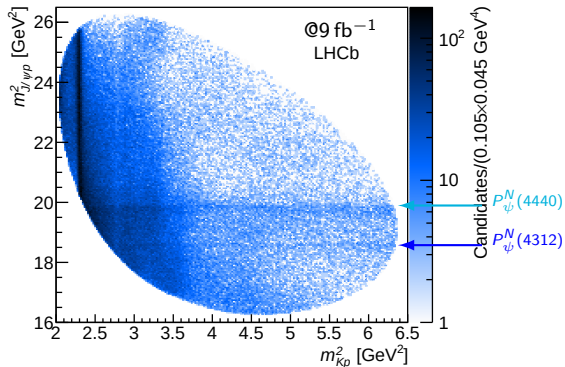
- Close to the $\Sigma_c \bar{D}^{(*)}$ threshold,
- Multiplicity matches spin combination:
 $1/2 \otimes 1 = 1/2 \oplus 3/2$
- Narrow(!): 10, 20, and 5 MeV for Γ_{BW}



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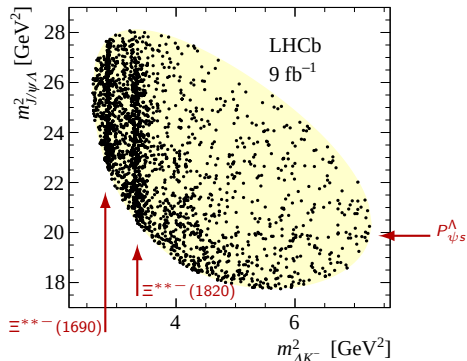
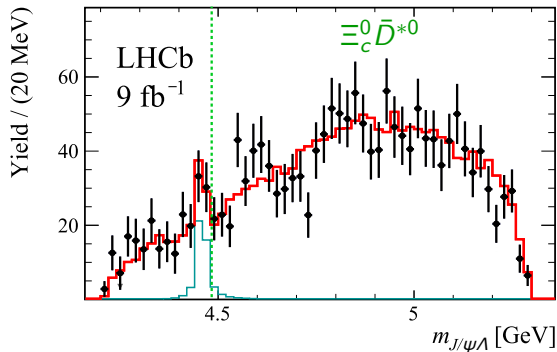


$$\Xi_b^- \rightarrow \underbrace{J/\psi \Lambda}_{p\Lambda_{\psi s}} K^-$$

Hint for the strange partners

$\Xi_b^- \rightarrow J/\psi(\rightarrow \mu^+\mu^-)\Lambda(\rightarrow p\pi^-)K^-$ data sample [Sci.Bull. 66 (2021) 1278-1287]

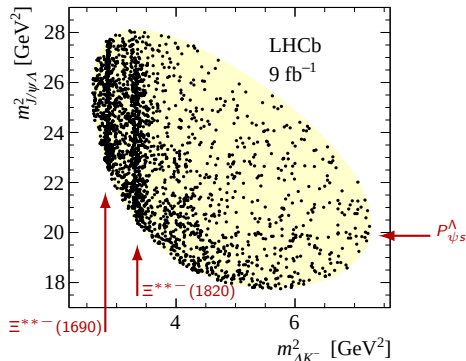
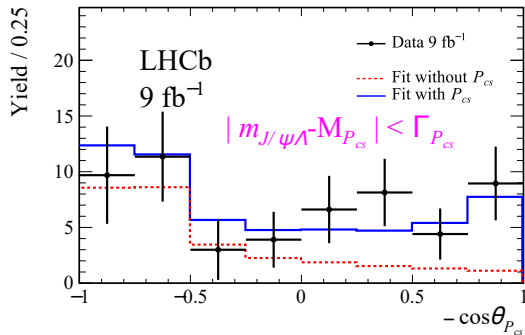
- Full data sample 1750 signals with purity 80%.
- The decay is dominated by the Ξ resonances
- $P_{\psi s}^\Lambda(4459)$: $m = 4458.8 \pm 2.9_{-1.1}^{+4.7}$ MeV, $\Gamma = 17.3 \pm 6.5_{-5.7}^{+8.0}$ MeV with 4.3σ significance



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$$B^- \rightarrow \underbrace{J/\psi \Lambda}_{P_{\psi s}^\Lambda} \bar{p}$$

$$B^- \rightarrow J/\psi(\rightarrow \mu^+ \mu^-) \Lambda(\rightarrow p \pi) \bar{p}$$

[LHCb-PAPER-2022-031 (in preparation)]

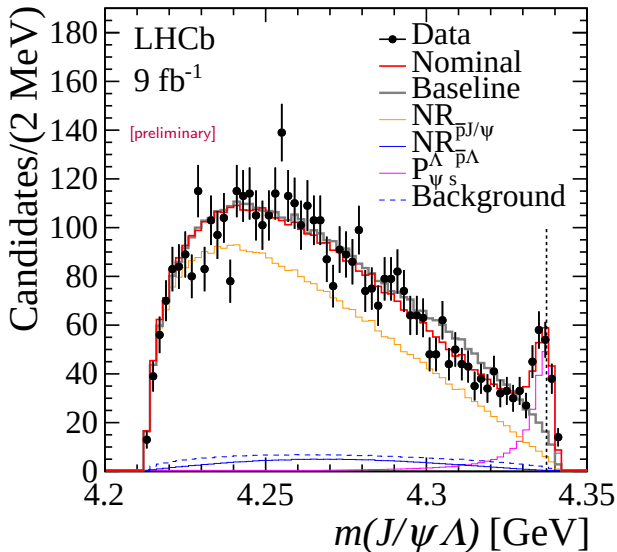
- Amplitudes:

- ▶ $\text{NR}(J/\psi p)$, $84.0 \pm 2.2\%$
- ▶ $\text{NR}(\Lambda \bar{p})$, $11.3 \pm 1.3\%$
- ▶ New $P_{\psi s}^\Lambda$, $12.5 \pm 0.7\%$,
- ▶ with parameters
 - ★ $m(P_{\psi s}^\Lambda) = 4338.2 \pm 0.7 \text{ MeV}$
 - ★ $\Gamma(P_{\psi s}^\Lambda) = 7.0 \pm 1.2 \text{ MeV}$

- $J^P = 1/2^-$ is preferred

- BW mass is close to $\Xi_c \bar{D}$ thresholds:

- ▶ 0.8 MeV above $\Xi_c^+ D^-$
- ▶ 2.9 MeV above $\Xi_c^0 \bar{D}^0$



What we know about Ω_c^{**} states

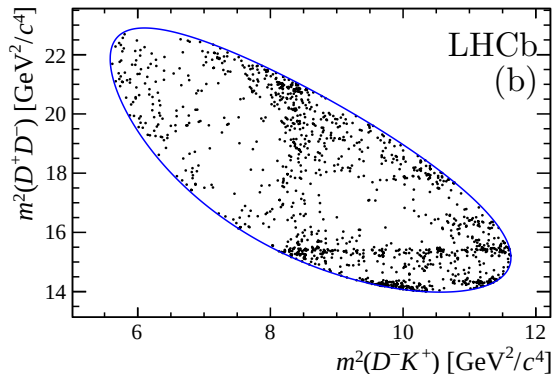
The threshold enhancement

The new higher states

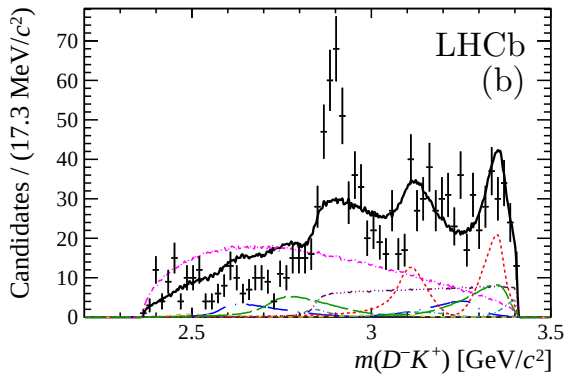
New Ξ_b states

$$B^+ \rightarrow D^+ \underbrace{D^- K^+}_{T_{cs}}$$

Dalitz plot for $B^+ \rightarrow D^+ D^- K^+$

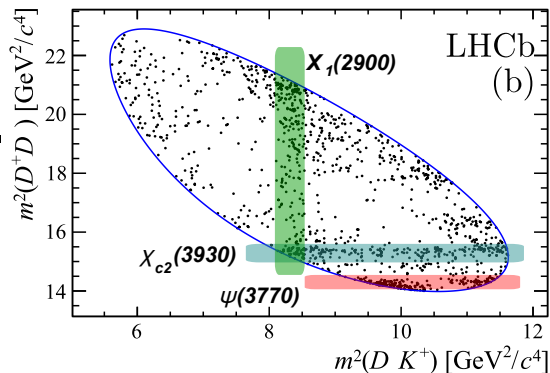


[LHCb, PRD102 (2020) 112003]

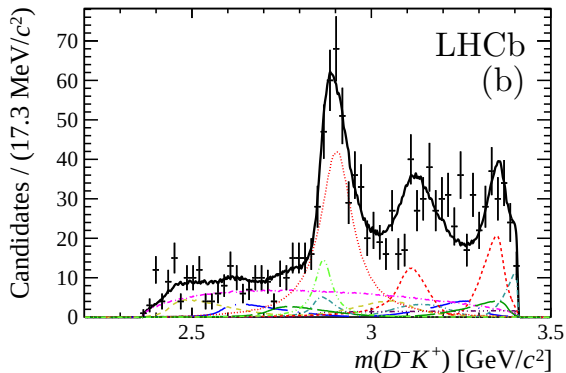


- Horizontal bands are resonances in $D^+ D^-$
- Hint for a vertical band around 8.5 GeV^2 in $m^2(D^- K^+)$
- Exotic candidate $T_{cs}(2900)$: $[\bar{c}\bar{s}ud]$
- Both quantum numbers $J^P = 0^+$ and 1^- are required in the fit

Dalitz plot for $B^+ \rightarrow D^+ D^- K^+$



[LHCb, PRD102 (2020) 112003]

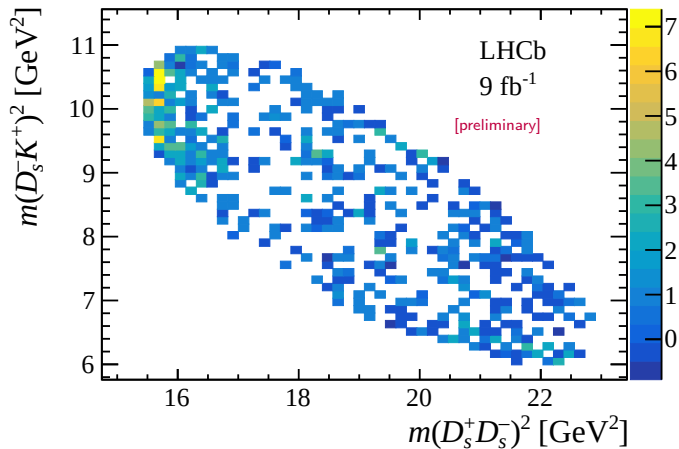


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$$B^+ \rightarrow \underbrace{D_s^+ D_s^-}_{\chi_{c0}/T_{\psi\phi}} K^+$$

Threshold enhancement at $D_s^+ D_s^-$ in $B^+ \rightarrow D_s^+ D_s^- K^+$ decays

[LHCb-PAPER-2022-018, 019 (in preparation)]



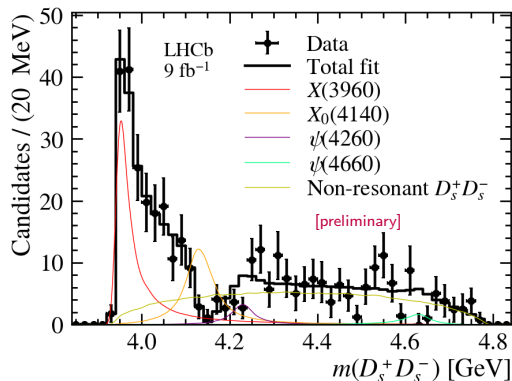
- $D_s^+ \rightarrow K^+ K^- \pi^+$
- Full data set, 9 fb⁻¹
- 360 signal candidates
- Purity of 84 %

$B^+ \rightarrow D_s^+ D_s^- K^+$ amplitude analysis

[LHCb-PAPER-2022-018, 019 (in preparation)]

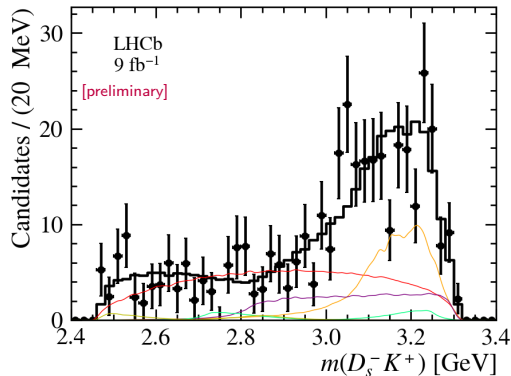
Main features of the data:

- Enhancement at the $D_s^+ D_s^-$ threshold
- Followed by a dip at 4.15 GeV.



Baseline model: $D_s^+ D_s^-$ resonances

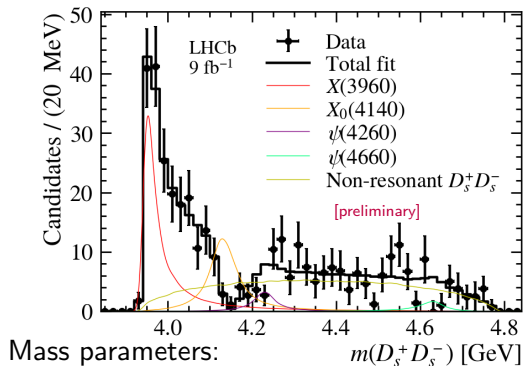
- 1^{--} : $\psi(4260) \sim 4\%$, $\psi(4660) \sim 2\%$
- 0^{++} : $X(3960) \sim 24\%$, $X(4140) \sim 18\%$, NR $\sim 50\%$



Alternative model using K -matrix

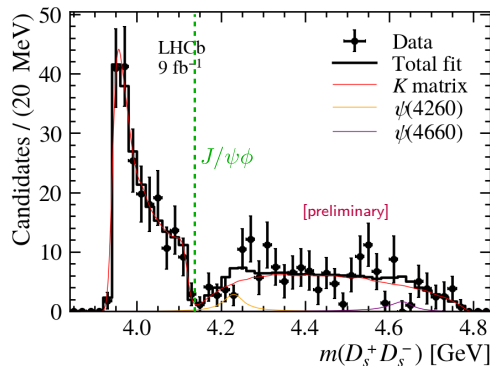
[LHCb-PAPER-2022-018, 019 (in preparation)]

Three interfering components in 0^{++} are replaced by the K -matrix.



Mass parameters:

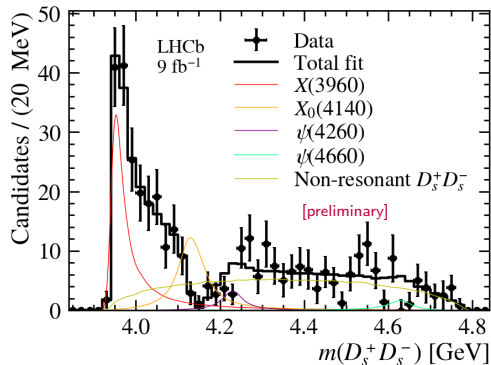
- 3960: $m = 3956 \pm 5 \pm 11 \text{ MeV}$,
 $\Gamma = 43 \pm 13 \pm 8 \text{ MeV}$
- 4140: $m = 4133 \pm 6 \pm 11 \text{ MeV}$,
 $\Gamma = 67 \pm 17 \pm 7 \text{ MeV}$



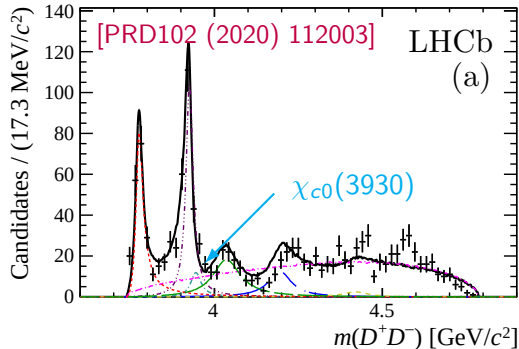
- Coupled channels $D_s^+ D_s^- / J/\psi \phi$
 - One K -matrix pole + bgd term
- Gives equally good fit

Is $X(3960)$ the same as $\chi_{c0}(3930)$ from D^+D^- ?

$B^+ \rightarrow (D_s^+ D_s^-) K^+$ by LHCb:



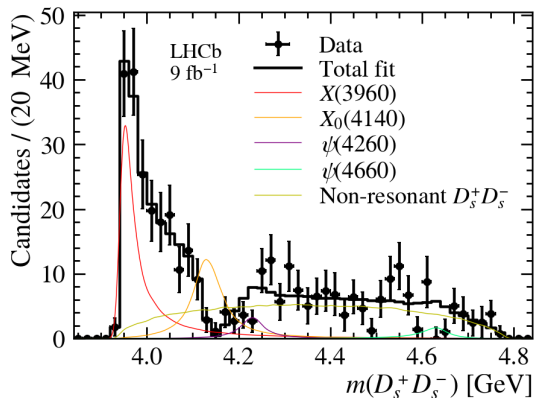
$B^+ \rightarrow (D^+ D^-) K^+$ by LHCb:



- Assuming to be the same, $\mathcal{B}(\chi_{c0} \rightarrow D^+ D^-) / \mathcal{B}(\chi_{c0} \rightarrow D_s^+ D_s^- P) \sim 0.3$
large molecular component, or large tetraquark component, $T_{\psi\phi}$
- [JHEP 06 (2021) 035] finds a state coupled to $D_s^+ D_s^-$ on the lattice

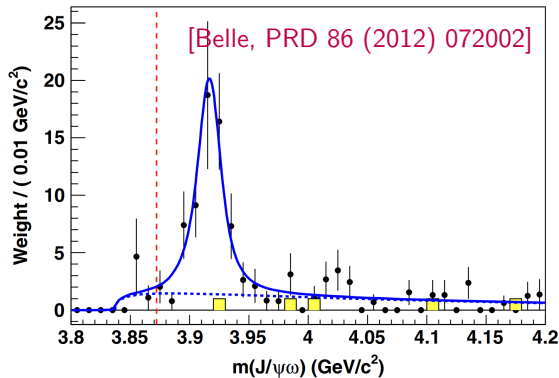
Is $X(3960)$ the same as $\chi_{c0}(3915)$?

$B^+ \rightarrow (D_s^+ D_s^-) K^+$ by LHCb:



[LHCb-PAPER-2022-018, 019 (in preparation)]

$\gamma\gamma \rightarrow J/\psi\omega$ by Belle:



- Belle sees a clean state in $J/\psi\omega$ with $J^P = 0^+$
- The $D_s^+ D_s^-$ signal might be a tail of the $\chi_{c0}(3915)$ state

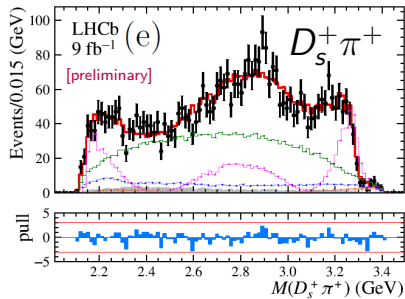
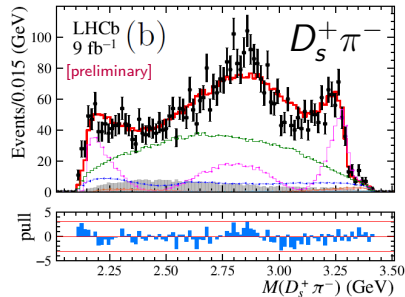
$$B^+ \rightarrow D^- \underbrace{D_s^+ \pi^+}_{T_{c\bar{s}}^a}$$

$$B^0 \rightarrow \bar{D}^0 \underbrace{D_s^+ \pi^-}_{T_{c\bar{s}}^a}$$

$T_{c\bar{s}}^a(2900)$ in the $D_s^\pm \pi^\pm$ system

[LHCb-PAPER-2022-026 (in preparation)]

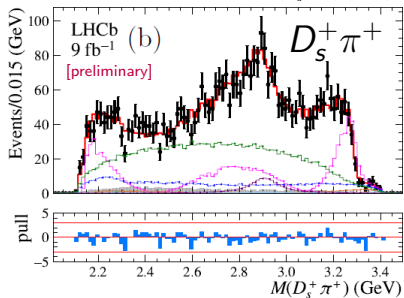
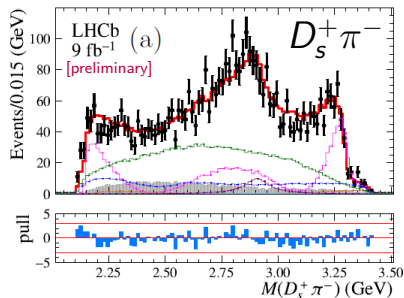
- 4420 B^0 decays and 3940 B^- decays, including charge-conjugated reactions
- Simultaneous fit using the isospin symmetry
- Main components in B^0/B^+ model:
 - ▶ $D^* \sim 17/14\%$
 - ▶ $D_2^* \sim 22/23\%$
 - ▶ $D\pi$ S-wave $\sim 45/48\%$.
- $T_{c\bar{s}}^a \sim 2\%$ needed ($> 5\sigma$), $J^P = 0^+$ is favored (7.5σ)
- Mass and width are close to these of $T_{c\bar{s}}^a(2900)$
 - ▶ $T_{c\bar{s}}^{a0}$: $m = 2892 \pm 14 \pm 15$ MeV,
 $\Gamma = 119 \pm 26 \pm 12$ MeV;
 - ▶ $T_{c\bar{s}}^{a++}$: $m = 2921 \pm 17 \pm 19$ MeV,
 $\Gamma = 137 \pm 32 \pm 14$ MeV



$T_{c\bar{s}}^a(2900)$ in the $D_s^\pm \pi^\pm$ system

[LHCb-PAPER-2022-026 (in preparation)]

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Summary and conclusion

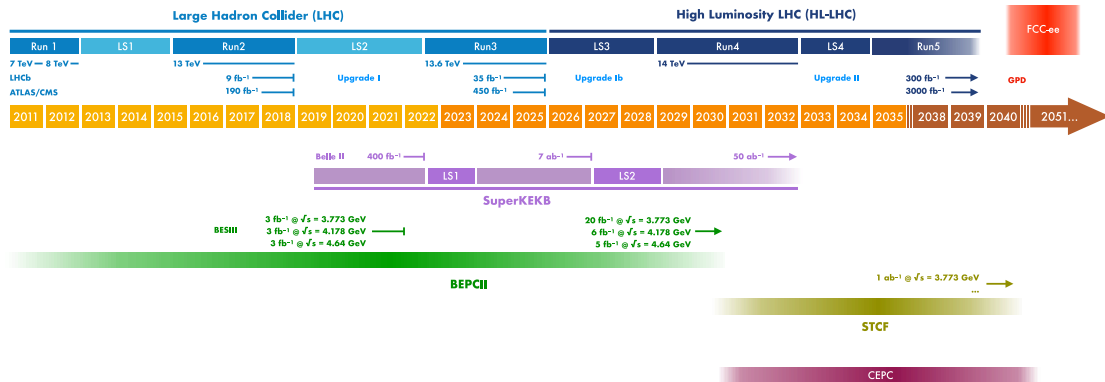
- With excellent performance of LHCb, we are exploring the uncharted waters
- This summer, more puzzle pieces for the quest of understanding nature complexity

Hints for the bigger picture:

- new $P_{\psi S}^{\Lambda}$ in B decays are close to $\Xi_c D$ threshold $\longleftrightarrow P_{\psi}^N$ are connected to $\Sigma_c \bar{D}^{(*)}$
Note, not the SU(3) flavour symmetry(!). Check $\Lambda_c \bar{D}^{(*)}$ and $\Xi'_c \bar{D}^{(*)}$ thresholds
- new $T_{c\bar{s}}^a$ appear to be similar to $T_{c\bar{s}}$
HQSS is in action? Where is the good/bad diquark mass difference?
- new $\chi_{c0}/T_{\psi\phi}$ close to the $D_s^+ D_s^-$ threshold.
A hadronic molecule or the scalar charmonium?

Updated timeline for LHC

[W. Altmannshofer, F. Archilli, arXiv:2206.11331]



LHCb:

- ramping up after major Upgrade I
- $\times 5$ statistics in Run 3(2023-2025) @13.6 TeV + Run 4(2029-2032) @14 TeV

Thank you for your attention